



Shri Gurm Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics & Telecommunication Engg.
In-Semester Examination - I
Academic Year: 2025-26 (semester -I)

Class: TY

**Course: Digital system Design
using Verilog (Elective)**

Course Code: PEC-EC305/PEC-EC313

Date: 18.08.2024

Time: 3 .00 PM to 3:45 PM

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		<u>Solve Any Two Question</u>	Marks	CO	BT
Q.1	a	Implement a parameterized N-bit Full adder using generate blocks in Verilog.	5	CO3	BT6
	b	Explain the concept of propagate and generate signals in Carry Look-Ahead Adders with the help of Truth table.	5	CO2	BT2
Q.2	a	Compare PLD, PLA and FPGA in terms of architecture and applications.	5	CO2	BT4
	b	Design a 4-bit binary subtractor using full subtractor modules in Verilog.	5	CO3	BT6
Q.3	a	What are step involved in software aspects of digital design,	5	CO1	BT1
	b	What is the role of test benches in Verilog/VHDL design?	5	CO1	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understanding Digital Design trends and software design aspects.
CO 2	Understanding Combinational and Sequential design aspects.
CO 3	Implementation of Combinational and Sequential Circuits using Verilog HDL.
CO 4	Understanding Data path and Control Path design aspects.
CO 5	Implementation of Data path and Control Path using Verilog HDL.



SGGS Institute of Engineering and Technology, Vishnupuri,
Nanded

Electronics and Telecommunication Engineering
In-Semester Examination (ISE)-1

Academic Year: 2025-26 (Semster-I)

T.Y. B. Tech.

Date: 20/08/2025

Note:

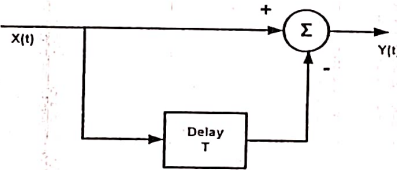
Analog Communication

Time: 15.00 - 15.45

(PCC-EC305)

Max. Marks: 20

1. Attempt all questions
2. Assume suitable data if necessary.
3. Use of a non-programmable calculator is permitted.
4. This paper contains 02 page.

Q.N.	Question	Marks	CO	BT
Q.1	A random variable has a probability density $f_X(x) = \begin{cases} (5/4)(1 - x^4), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere in } x \end{cases}$ Find a. $E[X]$ b. $E[4X + 2]$ and c. $E[X^2]$	[06]	PCC-EC-303.1	BT3
Q.2	Suppose that a Wide Sense Stationary (WSS) random process $X(t)$ with power spectrum $S_{XX}(\omega)$ is the input to the filter shown in Figure. Find the power spectrum of the output process $Y(t)$. 	[04]	PCC-EC-303.2	BT2
Q.3	Consider a random process $X(t)$ given by $X(t) = A \cos(\omega t + \theta)$ where A and ω are constants and θ is uniform random variable over $[0, 2\pi]$. Show that $X(t)$ is WSS.	[04]	PCC-EC-303.4	BT2
Q.4	A random variable X is uniformly distributed on the interval $(-5, 15)$. Another random variable $Y = e^{-X/5}$ is founded. Find a. $E[Y]$ b. $E[Y^2]$ c. σ_Y^2	[06]	PCC-EC-303.4	BT2

Course Outcomes (CO):

1. PCC-EC-303.1 – Understand the mathematical modelling of noise.



SGGS Institute of Engineering and Technology, Vishnupuri,
Nanded

Electronics and Telecommunication Engineering
In-Semester Examination (ISE)-1

Academic Year: 2025-26 (Semster-I)

T.Y. B. Tech.

Communication Systems Theory

(PCC-EC303)

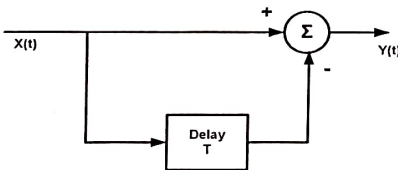
Date: 20/08/2025

Time: 15.00 15.45

Max. Marks: 20

Note:

1. Attempt all questions
2. Assume suitable data if necessary.
3. Use of a non-programmable calculator is permitted.
4. This paper contains 02 page.

Q.N.	Question	Marks	CO	BT
Q.1	A random variable has a probability density $f_X(x) = \begin{cases} (5/4)(1 - x^4), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere in } x \end{cases}$ Find a. $E[X]$ b. $E[4X + 2]$ and c. $E[X^2]$	[06]	PCC-EC-303.1	BT3
Q.2	Suppose that a Wide Sense Stationary (WSS) random process $X(t)$ with power spectrum $S_{XX}(\omega)$ is the input to the filter shown in Figure. Find the power spectrum of the output process $Y(t)$. 	[04]	PCC-EC-303.2	BT2
Q.3	Consider a random process $X(t)$ given by $X(t) = A \cos(\omega t + \theta)$ where A and ω are constants and θ is uniform random variable over $[0, 2\pi]$. Show that $X(t)$ is WSS.	[04]	PCC-EC-303.4	BT2
Q.4	A random variable X is uniformly distributed on the interval $(-5, 15)$. Another random variable $Y = e^{-X/5}$ is founded. Find a. $E[Y]$ b. $E[Y^2]$ c. σ_Y^2	[06]	PCC-EC-303.4	BT2

Course Outcomes (CO):

1. PCC-EC-303.1 – Understand the mathematical modelling of noise.

2. PCC-EC-303.2 – Analyze the behavior of analog and digital modulation systems in presence of noise.
3. PCC-EC-303.3 – Analyze the effect of channel noise on digital communication system.
4. PCC-EC-303.4 – Design the optimum filter like Matched filter to optimize the detector performance.
5. PCC-EC-303.5 – Understand the effect of inter symbol interference (ISI) in channel modelling.

SGGS Institute of Engineering and Technology, Vishnupuri, Nanded. ISE-1 Examination, Academic Year 2025-26 (Sem1)
Class: Third Year (Electronics and Telecommunication Engineering)

Date: 23/08/25	Subject: Microcontrollers and Applications, Subject code: PCC-EC 301
Time: 15 to 15:45 p.m.	Max. Marks : 20 , Duration : 45 minutes

Note: I. Use of calculator is not allowed II. All questions are compulsory to solve.

Sr. No.	Questions	CO	Marks
Q.1	State difference between microprocessor and microcontroller? Why is the crystal oscillator frequency of 8051 microcontroller based system is 11.0592 MHZ? Write 8051 assembly language program to load the accumulator with the value FFH and complement the Accumulator 100 times.	CO1	05
Q.2	Discuss all registers of 8051 Microcontroller. Show the contents of PSW after execution of the following instructions, MOV A, #0BFH ADD A, #2BH	CO2	05
Q.3	Explain PUSH, DA, DJNZ and MOVX instructions.	CO1	05
Q.4	Write a 8051 assembly language program to add the values F1H, C1H, FFH, 10H and 10H. Explain direct and indirect addressing mode of 8051 Microcontroller.	CO2	05
	END		
CO1: Interpret architecture of 8-bit microprocessor and microcontrollers.CO2: Develop program in 8051 in assembly language for the given operation.CO3: Develop program using timers and interrupts.CO4: Interface the memory and I/O peripherals to 8051 microcontrollers.CO5: Design microcontroller-based applications.			

Q.2	I. For two independent events A and B, prove that i. $P(A B) = P(A)$ ii. $P(B A) = P(B)$ II. Draw Venn Diagrams to show that $(A^c)^c = A$ III. Write formula for Permutation and Combination computing.	05	1, 2, 3	1, 2, 4
Q.3	Explain the Classification of Variables.	05	1, 2	2
Q.4	Explain the following Data Visualization types with Suitable Examples. i) Bar chart ii) Histogram iii) Pie-chart	05	3	3

*****!!!!!!All the Best!!!!!!*****

Details below are for Information only:

Bloom's Taxonomy: BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create
COs: Course Outcomes

CO1: Understand the basic knowledge on fundamental probability concepts, including random variable, probability of an event, additive rules and conditional probability

CO2: Apply the central limit theorem to sampling distribution and use estimation technique to determine point estimates confidence interval and sample size.

CO3: Understand several well-known distributions, including Binomial, Geometrical, Negative Binomial, Normal and Exponential Distribution.

Date: 02/02/25

Subject: Microcontrollers and Applications, Subject code: PCC-EC 206
Max. Marks : 20 , Duration : 20 minutes

Time: 14:40 to 15:00 p.m.

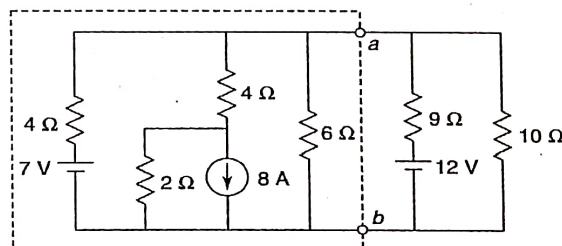
Note: I. Use of calculator is not allowed

II. All questions are compulsory to solve.

Sr. No.	Questions	CO	Marks
Q.1	State difference between microprocessor and microcontroller? Why is the crystal oscillator frequency of 8051 microcontroller based system is 11.0592 MHZ? Write 8051 assembly language program to load the accumulator with the value FFH and complement the Accumulator 100 times.	CO1	05
Q.2	Discuss all registers of 8051 Microcontroller. Show the contents of PSW after execution of the following instructions, MOV A, #0BFH ADD A, #2BH	CO2	05
Q.3	Explain PUSH, DA, DJNZ and MOVX instructionsMicrocontroller.	CO1	05
Q.4	Write a 8051 assembly language program to add the values F1H, C1H, FFH, 10H and 10H. Explain direct and indirect addressing mode of 8051	CO2	05
	END		
CO1: Interpret architecture of 8-bit microprocessor and microcontrollers.CO2: Develop program in 8051 in assembly language for the given operation.CO3: Develop program using timers and interrupts.CO4: Interface the memory and I/O peripherals to 8051 microcontrollers.CO5: Design microcontroller-based applications.			

OR

Find Norton's equivalent circuit for the portion of network shown in Fig. to the left of ab. Hence obtain the current in the 10 Ω resistor.



7

CO2

BT4